Q_1

(i)

Frame transmission time = 31 − 5.6 ns

= 25.4 ns

Maximum propagation time = 18.4 − 5.6

= 12.8 ns

(ii)

∴ Minimum Frame size = Bandwidth × 2 × Max. Prop. time

$= 10 \times 10^9 \times 2 \times 12.8 \times 10^{-9}$

= 256 bits

∴ For given frame, its size is $B \times T_{fr}$ [B = Bandwidth]

$= B \times 25.4$

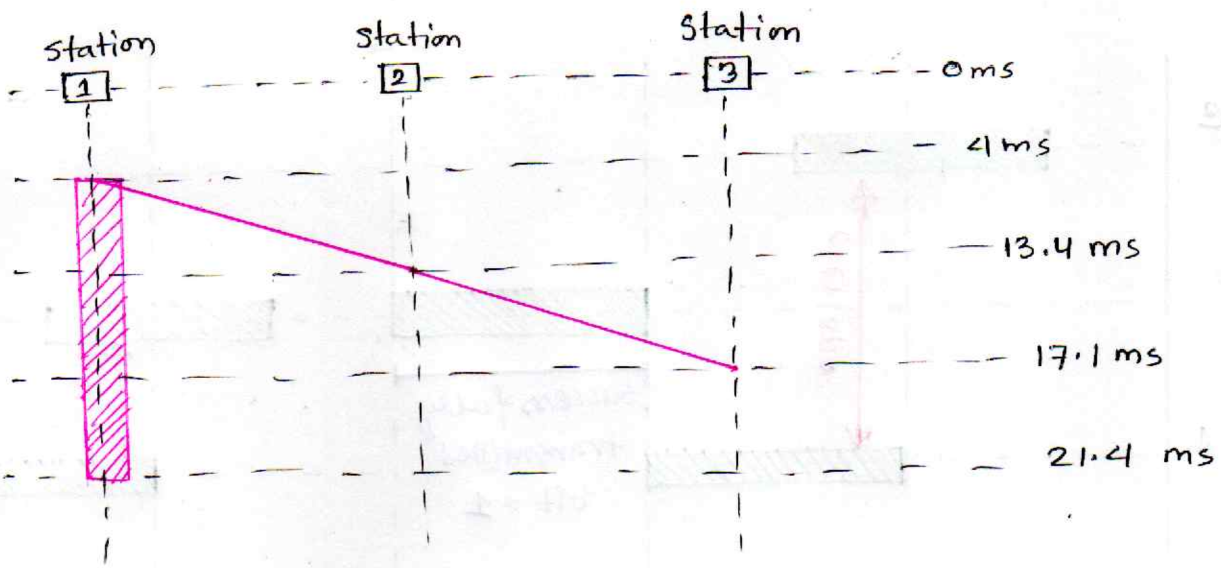
$= 10 \times 10^9 \times 25.4 \times 10^{-9}$

= 254 bits.

∴ The frame of image do not meet the threshold for minimum frame size. Ans

Q2

(I)



(II)

Transmission time = 21.4 − 4

= 17.4 ms

Max Propagation Time = 17.1 − 4 = 13.1 ms

(III)

Minimum Frame Size = Bandwidth × 2 × Max Propa. Time

$= 10^9 \times 2 \times 2 \times 13.1 \times 10^{-6}$

$= 52.4 \times 10^3$ bits

Frame size as per the diagram = Bandwidth × transmissin time

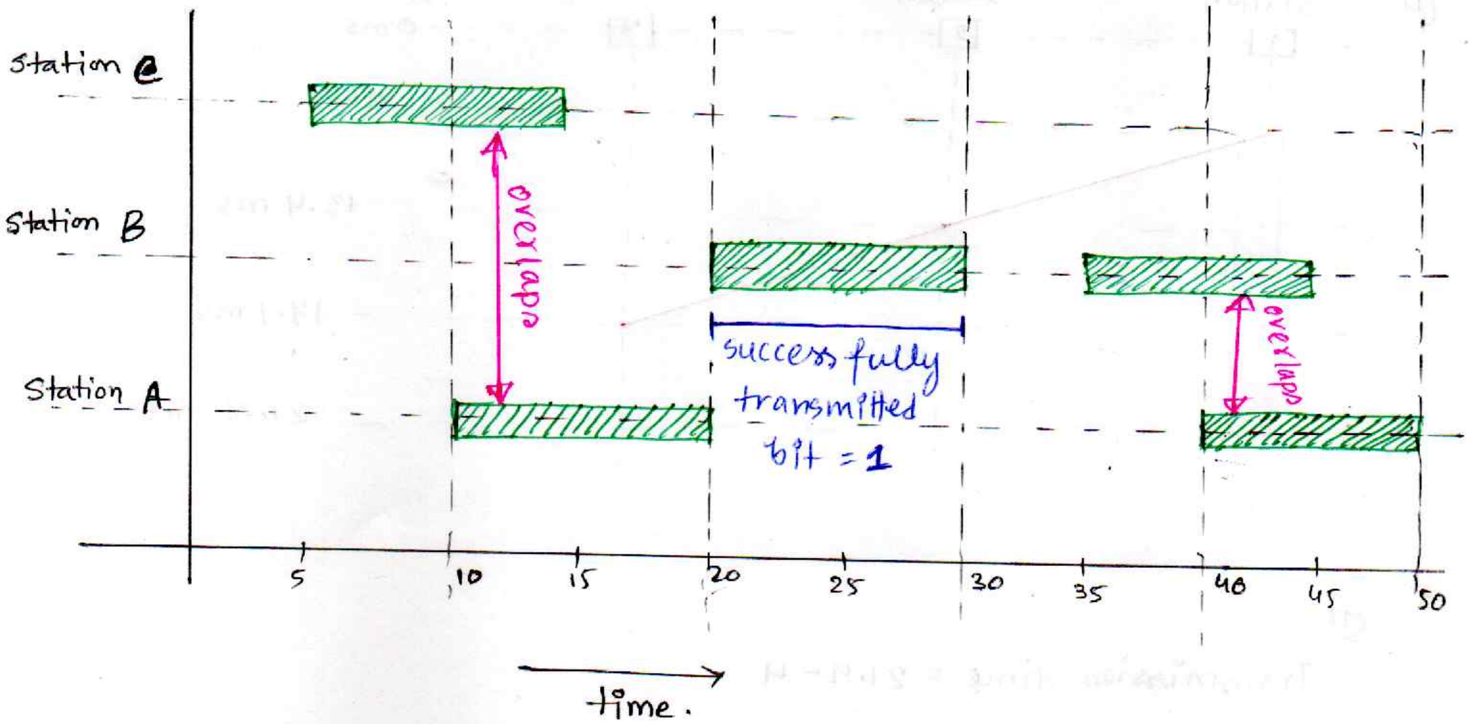
$= 10^9 \times 2 \times 17.4 \times 10^{-6}$

$= 34.8 \times 10^3$ bits

$\therefore 34.8 \times 10^3 < 52.4 \times 10^3$

∴ Minimum size/requirement not meet. Ans

Q5



Q4

(1) Total Distance = 2.4 + 3.6 + 2 = 8 m

$\therefore$ Propagation Time = $\frac{\text{Total Distance}}{\text{Propagation Speed}} = \frac{8}{2 \times 10^6} = 4 \times 10^{-6}$ s.

$\therefore$ Minimum Frame size = Bandwidth × 2 × Propagation Time

$= 2 \times 10^6 \times 4 \times 10^{-6} \times 2$

$= 16$ bits

⇒ If the frame size is smaller, it can't be detected clearly by other senders. Without facing collision, it may send and travel back.

(II)

$B \to D = 3.6 + 2 = 5.6\ m.$

$$t_{B \to D} = \frac{5.6}{2 \times 10^{6}} = 2.8 \times 10^{6}\ sec.$$

$\therefore$ B's signal reaches as $D = \cancel{2\mu s} + 2.8 \times 10^{6} s$

$= 4.8\ \mu s$

$\therefore$ Given,

D starts sending at $4.5\ \mu s$, without hearing from B. So, their transmission overlaps, resulting collission